

STACK QNO:7

QUESTION:

Question description

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output -1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or -1 , if $G(F(i))$ does not exist.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
8	8
3 7 1 7 8 4 5 2	31 71 11 17 81 14 51 21
EXPECTED OUTPUT	EXPECTED OUTPUT
1 4 4 4 -1 2 -1 -1	11 14 14 14 -1 21 -1 -1

Mandatory Test Cases

Test Case 1	Test Case 2
KEYWORD	KEYWORD
<code>if(a[j]>a[i])</code>	<code>for(i=0;i<n;i++)</code>

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
9	225	30

Source Code:

```
#include <stdio.h>
long long a[3005];
int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }
    int i;
    for(i=0;i<n;i++) {
        scanf("%lld", &a[i]);
    }

    for(i=0;i<n;i++) {
        int F_idx = -1;
        for (int j = i + 1; j < n; j++) {
            if(a[j]>a[i]) {
                F_idx = j;
                break;
            }
        }
        long long ans = -1;
        if (F_idx != -1) {
            for (int k = F_idx + 1; k < n; k++) {
                if (a[k] < a[F_idx]) {
                    ans = a[k];
                    break;
                }
            }
        }

        printf("%lld", ans);
        if (i < n - 1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```

Output:

```
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1
```

```
8
31 71 11 17 81 14 51 21
11 14 14 14 -1 21 -1 -1
```